

Cheat sheet — Markdown, Quarto and equations

IATDMCT Congress · Introduction to Markdown

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Keep this next to your keyboard. Everything here is written with the syntax it describes, so you can copy the source straight out of `cheatsheet.qmd`.

1 Markdown in one table

Purpose	Type this	Looks like
Heading	<code>## Section</code>	Section heading
Bold / italic	<code>bold / <i>italic</i></code>	bold / <i>italic</i>
Strikethrough	<code>~~retracted~~</code>	retracted
Inline code	<code>`CL = D/AUC`</code>	CL = D/AUC
Bulleted list	<code>- item</code>	• item
Numbered list	<code>1. item</code>	1. item
Task list	<code>- [x] done</code>	done
Link	<code>[text] (https://url)</code>	text
Image	<code>![Caption] (img.png)</code>	An image with caption
Blockquote	<code>> quoted</code>	inset quote
Footnote	<code>Text. [^a]</code>	superscript marker
Line break	two trailing spaces	new line, same paragraph
Divider	<code>---</code>	horizontal rule
Escape	<code>* _ \# \\$ \%</code>	literal characters

2 Quarto extras

Purpose	Syntax
YAML header	--- ... --- at the very top of the file
Callout	::: {.callout-note} ... ::: (also tip, warning, important, caution)
Columns	::: {.columns} + ::: {.column width="50%"}
Table caption + label	: Caption {#tbl-name}
Figure label	![Caption](img.png){#fig-name}
Equation label	\$\$... \$\$ {#eq-name}
Cross-reference	@sec-name, @fig-name, @tbl-name, @eq-name
Citation	[@key], bibliography in the YAML
Code chunk	```{r} ... ``` , options via # label:
Chunk without running	```r (no braces)
Raw HTML fallback	text

3 Equations — the essentials

Type this	You get	Type this	You get
x	inline x	x	display equation
$C_{\max}$	$C_{\max}$	C^2, x_i^2	C^2, x_i^2
$\frac{a}{b}$	$\frac{a}{b}$	$\frac{a}{b}$ (taller)	$\frac{a}{b}$ (taller)
$\sqrt{x}$	$\sqrt{x}$	$\sqrt[3]{x}$	$\sqrt[3]{x}$
$\ln, \exp, \max$	ln, exp, max	$\log_{10}$	$\log_{10}$
$\sum_{i=1}^n$	$\sum_{i=1}^n$	$\prod_{i=1}^n$	$\prod_{i=1}^n$
$\int_0^\infty$	$\int_0^\infty$	$\lim_{x \rightarrow 0}$	$\lim_{x \rightarrow 0}$
$\alpha \beta \gamma$	$\alpha \beta \gamma$	$\mu \sigma \lambda_z$	$\mu \sigma \lambda_z$
$\Delta \partial \nabla$	$\Delta \partial \nabla$	$\infty \pm \approx$	$\infty \pm \approx$
$\times, \cdot$	$\times, \cdot$	$\geq \leq \neq$	$\geq \leq \neq$
$\left(\right)$	$()$	$\left \right $	$ $
$\mathrm{mg/L}$	mg/L	steady state	steady state
$\backslash, \backslash., \backslashquad$	space, thin space, wide space	$\hat{\theta}$	$\hat{\theta}$

Multi-line building blocks:

```

$$
\begin{aligned}
a &= b + c \\
&= d
\end{aligned}
\qquad
\begin{cases}
x & \leq T
\end{cases}

```

```

y & t > T
\end{cases}
\quad
\begin{pmatrix} a & b \\ c & d \end{pmatrix}
$$

```

$$a = b + c \quad \begin{cases} x & t \leq T \\ y & t > T \end{cases} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Spacing, which is what makes an equation readable:

Type this	You get	What it does
<code>a\,b</code>	$a\,b$	thin space
<code>a\ b</code>	$a\ b$	normal space (units)
<code>a \quad b</code>	$a \quad b$	wide space
<code>\ln\!\left(x\right)</code>	$\ln(x)$	pull together with \!
<code>\operatorname{CL}</code>	CL	upright multi-letter label

4 Equations you will type most often

Copy the source column straight into a document. `$$ ... $$` on one line is display math; `$ ... $` is inline math.

Quantity	Source
First-order decay	<code>\$\$C(t) = C_0 e^{-k_e t}\$\$</code>
IV bolus, one compartment	<code>\$\$C(t) = \frac{D}{V_d} e^{-k_e t}\$\$</code>
Elimination rate constant	<code>\$\$k_e = \frac{CL}{V_d}\$\$</code>
Half-life	<code>\$\$t_{1/2} = \frac{\ln 2}{k_e}\$\$</code>
Clearance	<code>\$\$CL = \frac{D}{AUC}\$\$</code>
Volume of distribution	<code>\$\$V_d = \frac{D}{C_0}\$\$</code>
AUC after an IV bolus	<code>\$\$AUC_{0 \to \infty} = \frac{C_0}{k_e} = \frac{D}{CL}\$\$</code>
Trapezoidal AUC	<code>\$\$AUC_{0 \to t_n} = \sum_{i=1}^{n-1} \frac{C_i + C_{i+1}}{2} \Delta t_i\$\$</code>
AUC extrapolated to infinity	<code>\$\$AUC_{0 \to \infty} = AUC_{0 \to t_{last}} + \frac{C_{last}}{k_e}\$\$</code>
Mean residence time	<code>\$\$MRT = \frac{AUMC}{AUC}\$\$</code>
Steady-state volume	<code>\$\$V_{ss} = MRT \cdot CL\$\$</code>
Steady state, infusion	<code>\$\$C_{ss} = \frac{R_0}{CL}\$\$</code>
Approach to steady state	<code>\$\$f_{ss}(t) = 1 - e^{-k_e t}\$\$</code>
Loading dose	<code>\$\$D_L = \frac{C_{target}}{V_d} S\$\$</code>
Trough at steady state	<code>\$\$C_{min,ss} = \frac{F D}{V_d} \frac{e^{-k_e \tau}}{1 - e^{-k_e \tau}}\$\$</code>
Oral (Bateman) profile	<code>\$\$C(t) = \frac{F D k_a}{V_d (k_a - k_e)} \left(e^{-k_e t} - e^{-k_a t} \right)\$\$</code>
Time to peak, oral	<code>\$\$t_{max} = \frac{\ln(k_a / k_e)}{k_a - k_e}\$\$</code>

Quantity	Source
Peak after an oral dose	$C_{\max} = \frac{F D}{V_d} \left(\frac{k_a}{k_a - k_e} \right) \left(e^{-k_e t_{\max}} - e^{-k_a t_{\max}} \right)$
Bioavailability	$F = \frac{AUC_{\text{oral}}}{CL \cdot D_{\text{oral}}}$
Cockcroft–Gault	$CL_{\text{CR}} = \frac{(140 - \text{age})}{WT} \cdot S_{\text{Cr}} \cdot \begin{cases} 1 & \text{if male} \\ 0.85 & \text{if female} \end{cases}$
Dose adjustment	$D_{\text{adj}} = D_{\text{std}} \cdot \frac{CL_{\text{CR,pt}}}{CL_{\text{CR,ref}}}$
Weight-based clearance	$CL_i = CL_{\text{pop}} \left(\frac{WT_i}{70} \right)^{0.75} e^{\eta_i}$
Proportional residual error	$C_{ij} = C_{\text{pred},ij} \cdot (1 + \epsilon_{ij})$
Between-subject variability	$CV\% = \sqrt{e^{\omega^2} - 1} \cdot 100\%$
Michaelis–Menten	$-\frac{dC}{dt} = \frac{V_{\max} C}{K_m + C}$
Target attainment	$\frac{AUC_{24}}{MIC} \geq 400$
Bayesian posterior	$p(\theta \mid y) \propto p(y \mid \theta) \cdot p(\theta)$
MAP objective	$\hat{\theta} = \arg \max_{\theta} \left[-\sum_{j=1}^m \left(-\frac{1}{2} \left(\frac{C_j^{\text{obs}} - C_j^{\text{pred}}(\theta)}{\sigma_j} \right)^2 \right) \right]$

Rendered, so you can check your output at a glance:

$$C(t) = \frac{D}{V_d} e^{-k_e t} \quad t_{1/2} = \frac{\ln 2}{k_e} \quad CL = \frac{D}{AUC}$$

$$AUC_{0 \rightarrow t_n} = \sum_{i=1}^{n-1} \frac{C_i + C_{i+1}}{2} \Delta t_i \quad C_{ss} = \frac{R_0}{CL} \quad -\frac{dC}{dt} = \frac{V_{\max} C}{K_m + C}$$

5 Punctuation that bites

- `_` and `*` are formatting characters: write `AUC\MIC`, not `AUC_MIC`.
- `%` must be `\%` inside math if you ever export to PDF.
- `$` in prose: currency is `\$5`, and an unmatched `$` swallows the rest of the paragraph.
- Units belong in `\mathrm{}`. Multi-letter variables too: `\mathrm{AUC}`, `\mathrm{CL}`.
- Functions are commands: `\ln(2)`, `\exp(x)`, `\max_i`.
- Subscripts with more than one character need braces: `C_{\max}`, `t_{1/2}`.
- Display math wants its own paragraph — blank line before and after.
- `\left` and `\right` are a pair; a missing one breaks the equation.
- Prefer `\mu`, `\beta`, `\alpha` to typed Unicode when the document may become a PDF.

6 Command line

```

quarto preview file.qmd      # live preview while you write
quarto render file.qmd       # build the file next to its source
quarto render                # build the whole project (uses _quarto.yml)
quarto render file.qmd --to docx # a different output format
quarto render file.qmd --to typst
quarto check                 # is the toolchain complete?
quarto install tinytex       # LaTeX for PDF output

```

Settings that matter for a talk:

```

format:
  beamer:
    slide-level: 2           # `##` starts a slide; without it the deck collapses
    aspectratio: 169
    header-includes: |       # footer with page numbers
      \setbeamertemplate{footline}{%
        \hspace*{2ex}Your Name · IATDMCT Congress%
        \hfill\insertframenumber{}/\inserttotalframenumber\hspace*{2ex}%
      }

```

`{.smaller}`, `{.incremental}` and background colours are reveal.js features and are ignored in a PDF — fit a slide by cutting words.